Pergamon

Nonlinear Analysis, Theory, Methods & Applications, Vol. 30, No. 8, pp. 5303–5314, 1997
Proc. 2nd World Congress of Nonlinear Analysts
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0362-546X/97 $17.00 + 0.00

PII: S0362-546X(96)00149-6

TAKENS EMBEDDING THEOREMS FOR FORCED AND STOCHASTIC SYSTEMS

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1. INTRODUCTION

Takens Embedding Theorem [1,2,3] provides the theoretical foundation for the analysis of time series generated by nonlinear deterministic dynamical systems. Informally, it says that if we take a scalar observable φ of the state x of a deterministic dynamical system then *typically* we can reconstruct a copy of the original system by considering blocks $(\varphi(x_t), \varphi(x_{t+\tau}), \varphi(x_{t+2\tau}), \ldots, \varphi(x_{t+(d-1)\tau}))$ of d successive observations of φ, for d sufficiently large. Here x_t is the state of the system at time t, and $\tau > 0$ is some sampling interval. Since x_t will usually be unknown, whilst $\varphi(x_t)$ is a quantity we can measure in practice, this result has stimulated a vast range of applications in fields ranging from fluid dynamics, through electrical engineering to biology, medicine and economics (for a good overview see e.g. [4]). In particular, it has led to both a re-examination of old data sets, and to the construction of new experiments, with the aim of detecting and perhaps even taking advantage of deterministic behaviour in time series which were previously thought to be random. One might even say that this one theorem has given rise to virtually a new branch of nonlinear dynamics, often informally called *chaotic time series analysis.*

However, for Takens Theorem to be valid, we need to assume that both the dynamics is deterministic (*i.e.* that there is some mapping f such that $x_{t+\tau} = f(x_t)$) and that both the dynamics, and the observations are autonomous (so that f and φ depend on x only). Unfortunately, these assumptions fail to hold in many applications. Thus, most real world systems are noisy to at least some extent. Such noise may appear in both the dynamics and the observations, in either case the existing form of Takens Theorem is not applicable. The inability of Takens Theorem to rigorously deal with stochasticity has been a severe drawback, and is perhaps one reason why this approach to time series has received less attention from the traditional signal processing community than it deserves. Furthermore, even if the noise is sufficiently low for us to be able to ignore it, many deterministic systems are not autonomous. Thus, in the laboratory we often force an experimental system in order to elicit interesting behaviour, whilst in the case of naturally occurring systems it is very rare for us to be able to isolate the system to ensure that there are no external influences. Finally, many systems in the fields of control and communications are designed to process some arbitrary input and hence also cannot be treated as autonomous.

It is therefore highly desirable to extend Takens's framework both to deterministically forced systems and to stochastic systems. The aim of this paper is to describe recent progress in this direction, and in particular to discuss several generalizations of Takens Theorem which are applicable to these cases. It turns out that essentially the same approach can encompass a

number of different cases. In particular the theorems given below apply to deterministically forced systems, to a wide class of stochastic systems (which we think of as deterministic systems driven by some stochastic process), to input-output systems and to irregularly sampled systems.

2. THE STANDARD TAKENS THEOREM

In this section we outline the usual framework for studying time series generated by deterministic dynamical systems, give a statement of the standard Takens Theorem and briefly discuss some of its consequences. The behaviour of many real systems is often modelled by a deterministic finite dimensional dynamical system. By this we mean that the state of the system is described at any given instant by a point x lying on a m-dimensional manifold M which will often be some subset of $\mathbb{R}^k$. The time evolution of this state is described by a map $f^t: M\times\mathbb{R} \to M$ such that if the system is in state x at time t then it is in state $f^\tau(x)$ at time $t+\tau$. In most cases f^τ is given by the solution of some ordinary differential equation on M.

In many practical situations we do not have access to the state x and can merely observe some function $\varphi(x)$ of it. Here $\varphi: M \to \mathbb{R}$ is called the *measurement function*, it corresponds to measuring some observable property of the system such as position or temperature. The evolution of this quantity with time is then given by $\varphi(f^t(x_0))$. In practice we can only observe this at discrete time intervals. For simplicity we assume that these intervals are all the same so that in fact we observe the sequence $\varphi_n = \varphi(x_n)$ for $n = 1,2, \ldots$ where $x_n = f^{n\tau}(x_0)$ and $\tau > 0$ is the *sampling interval*. For obvious reasons $\{\varphi_n\}$ is called a *time series*. By rescaling time we may as well assume that $\tau = 1$, so that the sequence of states $\{x_n\}$ is given by the discrete dynamical system $x_{n+1} = f(x_n)$ where $f = f^1$ and f^n is just f composed n times.

The observed measurement φ_n is one dimensional whilst x_n in general lies in some higher dimensional space. At first sight it might thus appear that φ_n contains relatively little information about the behaviour of x_n and that the fact that φ_n originates in the deterministic process f^n is of little use. Remarkably, however, Takens Embedding Theorem[1] (see also [3,5,6,7]) shows that for typical f and φ it is possible to reconstruct f from the observed time series up to some (unknown) smooth co-ordinate change.

More precisely, fix some d (called the *embedding dimension*) and define the *delay embedding map* $\Phi_{f,\varphi}: M \to \mathbb{R}^d$ by

$$\Phi_{f,\varphi}(x) \quad = \quad (\varphi(x), \varphi(f(x)), \ldots, \varphi(f^{d-1}(x)))$$

Suppose that M is compact, let $\mathcal{D}^r(M)$ be the set of $\mathcal{C}^r$ diffeomorphisms of M, and $\mathcal{C}^r(M,\mathbb{R})$ the set of observation functions on M, both endowed with the $\mathcal{C}^r$ topology (see *e.g.* [8] for a definition). Recall that a subset $\mathcal{A} \subset \mathcal{D}^r(M)\times\mathcal{C}^r(M,\mathbb{R})$ is called *residual* (or a G_δ) if it contains a countable intersection of open dense sets. Since $\mathcal{D}^r(M)$ and $\mathcal{C}^r(M,\mathbb{R})$ are both Baire spaces, such a set is itself dense. A property which holds on a residual set is called *generic*, and this is one of the most commonly used notions of "typical" in the field of dynamical systems (though not the only one, *e.g.* see [3]). Finally, recall that a smooth map $\Psi: M \to N$ between manifolds M and N is called an *embedding* if it maps M diffeomorphically onto its image; if M is compact this is equivalent to both Ψ and its derivative $T_x\Psi$ being injective (*i.e.* 1-1) on the whole of M. Takens Embedding Theorem then states:

Theorem 1 (Takens [1]) Let M be a compact m dimensional manifold. If $d \geq 2m+1$, then the set of (f,φ) for which the map $\Phi_{f,\varphi}$ is an embedding is open and dense in $\mathcal{D}^r(M)\times\mathcal{C}^r(M,\mathbb{R})$ for $r \geq 1$.

Note that Takens first proved this for $r \geq 2$, but it is relatively easy to extend this to $r = 1$ (see [7]). A weaker version of this theorem was proved independently by Aeyels [2]. It turns out to be possible to give a simple characterization of the set of f for which $\Phi_{f,\varphi}$ is an embedding. In particular, as Huke [7] points out, Takens actually deduces the above theorem from the following "unstated Takens Theorem":

Theorem 2 Suppose that $f \in \mathcal{D}^r(M)$ has only a finite number of periodic orbits of period less than d, and the eigenvalues of each such periodic orbits are distinct. If $d \geq 2m+1$, then there is an open and dense set of $\varphi \in \mathcal{C}^r(M,\mathbb{R})$ for which $\Phi_{f,\varphi}$ is an embedding.

Note that by a simple generalization of the first part of the Kupka-Smale Theorem (*e.g.* [9]) maps f satisfying the conditions of this theorem are open and dense in $\mathcal{D}^r(M)$. The condition on the eigenvalues may seem a little strange, but is in fact almost a necessary condition for embedding. In particular it is easy to see that $f \in \mathcal{D}^r(M)$ has a fixed point $x \in M$ such that $T_x f$ has two linearly independent eigenvectors with the same eigenvalue then $\Phi_{f,\varphi}$ fails to be an immersion at x for all $\varphi \in \mathcal{C}^r(M,\mathbb{R})$. More generally, if x is a periodic point of period $q < d$, then $\Phi_{f,\varphi}$ fails to be an immersion at x if there are more than q linearly independent eigenvectors of $T_x f^q$ with the same eigenvalue.

Theorem 1 provides the justification for the majority of numerical methods used to analyse time series generated by nonlinear deterministic dynamical systems. It implies that for typical f and φ, the image $\Phi(M)$ of M under $\Phi = \Phi_{f,\varphi}$ is completely equivalent to M itself, apart from the smooth invertible change of co-ordinates given by Φ. Furthermore since Φ has a smooth inverse we can define the map $F = \Phi \circ f \circ \Phi^{-1}$ on $\Phi(M)$. Then F has the same dynamics as f, but seen in the new co-ordinates given by the co-ordinate change Φ. In particular, all the co-ordinate independent properties of F and f will be identical. This includes such features as the numbers and topological types of fixed points, periodic orbits and other invariant sets as well as such geometric invariants as the eigenvalues of fixed and periodic points, and the correlation dimension and Liapunov exponents of corresponding invariant measures (e.g. see [4,5,10,11]). Thus, in particular one can measure quantities such as the correlation dimension or the Liapunov exponents of F, and be certain that these are the same as those of the original (unknown) system f. The importance of this is that these invariants of F can be estimated directly from the observed time series $\{\varphi_n\}$. This is because if we define the point $z_n \in \mathbb{R}^d$ by the delay co-ordinates $z_n = (\varphi_n, \varphi_{n+1}, \ldots, \varphi_{n+d-1})$ then

$$\begin{aligned} z_n &= (\varphi(f^n(x_0)), \varphi(f^{n+1}(x_0)), \ldots, \varphi(f^{n+d-1}(x_0))) \\ &= (\varphi(x_n), \varphi(f(x_n)), \ldots, \varphi(f^{d-1}(x_n))) \\ &= \Phi(x_n) \end{aligned}$$

Hence z_n is in the image of Φ and so we can apply F to it:

$$\begin{aligned} F(z_n) &= \Phi \circ f \circ \Phi^{-1}(z_n) &&= \Phi \circ f \circ \Phi^{-1}(\Phi(x_n)) \\ &= \Phi \circ f(x_n) &&= \Phi(x_{n+1}) \\ &= z_{n+1} \end{aligned}$$

Thus the dynamics of F on $\Phi(M)$ simply consists of sliding d successive elements of the time series $\{\varphi_n\}$ along by one time step, i.e. $F(\varphi_n, \varphi_{n+1}, \ldots, \varphi_{n+d-1}) = (\varphi_{n+1}, \varphi_{n+2}, \ldots, \varphi_{n+d})$. As we noted above this is completely equivalent to the original dynamics $x_{n+1} = f(x_n)$ up to the co-ordinate change Φ, and all the co-ordinate invariant properties of F and f are identical. Yet, whilst the original dynamics was unobservable, the dynamics in terms of F and z_n is express-

ible purely in terms of the observed time series $\{\varphi_n\}$. Thus if we have a sample of $\{\varphi_n\}$, we can deduce many of the properties of F, and hence of f. Thus to summarize, Takens theorem allows us to reconstruct the unknown dynamical system which gave rise to the time series simply by constructing a new state space out of successive observations of the time series.

There are of course many practical obstacles which need to be surmounted if one is to use this in practice. Thus, for instance, we usually don't know m, and hence don't know what value of embedding dimension d we should use. Similarly the choice of sampling interval τ can significantly affect the performance of practical techniques. Such difficulties can however be overcome (e.g. see [4,11]) and today the estimation of fractal dimensions or Liapunov exponents from time series is fairly routine matter.

The existence of F also implies that the time series $\{\varphi_n\}$ is entirely deterministic; by iterating F we see that d successive values $\varphi_n, \varphi_{n+1}, \ldots, \varphi_{n+d-1}$ of the time series completely determine all its future values $\varphi_{n+d}, \varphi_{n+d+1}, \varphi_{n+d+2}, \ldots$. This suggests that it must be possible to forecast the future behaviour of $\{\varphi_n\}$. In particular, consider the map F. Its first d-1 components are trivial, since they consist of just shifting the argument of F by one time step. Denote the last component of F by $G : \Phi(M) \to \mathbb{R}$. Then $G(\varphi_n, \varphi_{n+1}, \ldots, \varphi_{n+d-1}) = \varphi_{n+d}$, or in other words G simply predicts the time series $\{\varphi_n\}$ one time step ahead! Repeating this process, we can in principle forecast $\varphi_{n+d+1}, \varphi_{n+d+2}, \ldots$, (though if f and hence F is chaotic then there is an upper limit on how far into the future we can predict in practice). Of course, we usually do not know G, but once again we can estimate it from a sample of the time series. Not only can this be useful in itself, but it also forms the basis of a variety of techniques for noise reduction, signal separation, control and synchronization (again see *e.g.* [4,11]).

Unfortunately, as already discussed in §1, Theorem 1 requires the dynamics to be both deterministic and autonomous, and this is not the case in many real applications. It is therefore important to develop generalizations of Takens Theorem applicable to stochastic systems on one hand, and forced systems on the other. We shall first consider the case of deterministic forcing, since this is somewhat easier to understand, but still involves most of the significant issues which arise in the case of stochastic systems.

3. DETERMINISTICALLY FORCED SYSTEMS

As before we shall assume that the state of the system under consideration is given by a point $x \in M$ and the dynamics is given by a smooth map f. However, we now assume that f is forced by some second system $g : N \to N$. As is usual, we study the forced dynamics of f by expanding the phase space and incorporating the forcing dynamics. This leads to a dynamical system on $M \times N$ of the form

$$\begin{aligned} x_{i+1} &= f(x_i, y_i) \\ y_{i+1} &= g(y_i) \end{aligned}$$

In the same way that in the unforced case f was required to be a diffeomorphism, we shall need the map $x \mapsto f(x,y)$ to be a diffeomorphism of M for every $y \in N$. We shall employ the notation $\mathcal{D}^r(M \times N, M)$ for the set of such f in $\mathcal{C}^r(M \times N, M)$. We shall also need g to be a diffeomorphism of N, which in turn means that the pair (f,g) is a diffeomorphism of $M \times N$. In line with common usage we shall call the pair (f,g) a *skew product* on $M \times N$.

A simple example of this is given by a periodically forced ordinary differential equation

$$\frac{dx}{dt} = \Psi(x,t)$$

where $\Psi(x,t)$ is a time dependent vector field which is periodic in time, so that $\Psi(x,t+T) = \Psi(x,t)$ for some period $T > 0$. Let f^s be the solution of the differential equation, so that $f^s(x,t)$ is the state of the system at time $t+s$, if the system was in state x at time t. Suppose that we have a measurement function $\varphi : M \to \mathbb{R}$ which we observe with a sampling interval $\tau > 0$. Then the ith observation in our time series is given by $\varphi(f^{i\tau}(x,0)) = \varphi(x_i)$ where x_i satisfies $x_{i+1} = f^\tau(x_i,t_i)$ with $t_{i+1} = t_i + \tau$. But $f^\tau(x_i,t_i) = f^\tau(x_i,t_i+T)$ and hence if we write $\theta_i = t_i/T$ we get the skew product on $M\times\mathbb{T}^1$, where $\mathbb{T}^1$ is the unit circle, given by

$$\begin{aligned} x_{i+1} &= f(x_i,\theta_i) \\ \theta_{i+1} &= \theta_i + \omega \qquad \pmod 1 \end{aligned}$$

where $f(x,\theta) = f^\tau(x,\theta T)$ and $\omega = \tau/T$. In this case, therefore, g is simply a rigid rotation of the circle through the angle ω. If the sampling interval τ is incommensurate to the period T then ω is irrational and the orbit of any θ is dense in $\mathbb{T}^1$, whilst if the sampling interval is rationally related to the forcing period then ω is rational and every orbit of g is periodic. In many circumstances, particularly within the context of a laboratory experiment one will of course sample at the forcing period, so that $\tau = T$ and g is the identity $g(\theta) = \theta$.

Returning to the case of a general forced system, given that (f,g) is a deterministic dynamical system on the manifold $M\times N$, it might at first sight seem that the standard Takens Theorem is in fact applicable to this situation, simply by considering the dynamics (f,g) and the observation function $\psi : M\times N \to \mathbb{R}$ given by $\psi(x,y) = \varphi(x)$. The problem with this idea is the word *generic* in the statement of Takens Theorem. Thus the standard Takens Theorem would allow us to conclude that generically the map $\Phi_{f,g,\psi} : M\times N \to \mathbb{R}^d$ is an embedding, where $\Phi_{f,g,\psi}$ is given by

$$\Phi_{f,g,\psi}(x,y) = (\psi(x,y), \psi((f,g)(x,y)), \psi((f,g)^2(x,y)), \ldots, \psi((f,g)^{d-1}(x,y)))$$

Unfortunately, the notion of typical that is given by Takens Theorem to us here is not the one that we usually want in practical applications. To explain this, let us first consider the example of periodically forced systems and for simplicity ignore perturbations in f and g and only consider the density of observation functions giving an embedding. Suppose that we attempted to apply the standard Takens Theorem to a periodically forced system, which by above has the form $x_{i+1} = f(x_i,t_i)$, $\theta_{i+1} = \theta_i + \omega \pmod 1$ with $x_i \in M$ and $\theta_i \in \mathbb{T}^1 = N$. We would then conclude that arbitrarily close to a given observation function $\psi_0(x,\theta) = \varphi_0(x)$ there was an observation function $\psi(x,\theta)$ for which $\Phi_{f,g,\psi}$ was an embedding. Unfortunately, typically such a ψ would have nontrivial dependence on θ. In other words, *our measurements would depend on the phase of the forcing.* In most practical applications, this will not be the case: *e.g.* if we measure the displacement of a forced pendulum, or a voltage in a forced electronic oscillator, we do not expect our measurement apparatus to give explicitly time dependent results. More formally, the standard Takens Theorem tells us that $\Phi_{f,g,\psi}$ is an embedding for generic $\psi \in \mathcal{C}^r(M\times\mathbb{T}^1,\mathbb{R})$, and generic functions in $\mathcal{C}^r(M\times\mathbb{T}^1,\mathbb{R})$ have explicit θ dependence. What we really want is $\Phi_{f,g,\psi}$ to be an embedding for generic φ in the space $\mathcal{C}^r(M,\mathbb{R})$ of ordinary time independent observation functions.

Similarly, in the case of a general driven system (f,g) the standard Takens Theorem says that $\Phi_{f,g,\psi}$ is an embedding for an open dense subset of $\mathcal{D}^r(M\times N)\times\mathcal{C}^r(M\times N,\mathbb{R})$, whilst we want it to be an embedding for an open dense set in $\mathcal{D}^r(M\times N,M)\times\mathcal{D}^r(N)\times\mathcal{C}^r(M,\mathbb{R})$. Since $\mathcal{C}^r(M,\mathbb{R})$ is not

generic in $\mathcal{C}^r(M\times N,\mathbb{R})$, we cannot conclude that typical skew products and typical functions on M lead to an embedding (note also that $\mathcal{D}^r(M\times N,M)\times\mathcal{D}^r(N)$ is not generic in $\mathcal{D}^r(M\times N)$ but this can be overcome by use of Theorem 2).

We thus conclude that the existing version of Takens Theorem is not particularly useful in the context of forced systems. Fortunately, it turns out to be possible to generalize the theorem to this case. In fact, we are able to give two versions of the generalized theorem, one relevant to the situation where the forcing is unknown, and the other to the situation where we have independent knowledge of the state y of the forcing system. This will for instance be the case if we control the forcing, as in many laboratory experiments. If this is so, then it is unnecessary to reconstruct the forcing dynamics, and as we shall see, this permits us to use a smaller embedding dimension.

Before we state these two theorems, we need to develop some notation. Since the observation functions $\varphi : M \to \mathbb{R}$ is a function of x only, the ith component of the delay map $\Phi_{f,g,\varphi} : M\times N \to \mathbb{R}^d$ depends only on the x co-ordinate of $(f,g)^{i-1}(x,y)$. It is convenient to write this as $f^{(i-1)}(x,y)$, where $f^{(i)} : M\times N \to M$ is defined by $f^{(i+1)}(x,y) = f(f^{(i)}(x,y),g^i(y))$ with $f^{(0)}(x,y) = x$. We can then write the delay map as

$$\Phi_{f,g,\varphi}(x,y) \quad = \quad (\varphi(f^{(0)}(x,y)), \varphi(f^{(1)}(x,y)), \ldots, \varphi(f^{(d-1)}(x,y)))$$

Ideally, we would like to conclude that if $d \geq 2(m+n) + 1$ then this is an embedding for generic f and generic φ, for any given g; in other words we might hope that there is sufficient freedom in perturbing f and φ to allow us to fix the forcing system. Unfortunately, this is not the case, and it turns out that we require the same conditions on g as Theorem 2 above imposed on f. We also require $\dim M \geq 1$. This is not much of a restriction, since if M is 0-dimensional and compact then it consists of a finite number of points, and hence if f: $M\times N \to M$ is continuous, then it must be independent of y, in other words the forcing plays no role. Thus $\mathcal{D}^r(M\times N,M)$ consists of a finite number of maps, and if $\Phi_{f,g,\varphi}$ was an embedding for a dense set of f, then it would be an embedding for all f, which in general is not possible. If we exclude these two problems we get the following theorem:

Theorem 3 (Forced Takens Theorem, Stark [12]) Let M and N be compact manifolds of dimension $m \geq 1$ and n respectively. Suppose that $d \geq 2(m+n) +1$ and the periodic orbits of period $\leq d$ of $g \in \mathcal{D}^r(N)$ are isolated and have distinct eigenvalues. Then for $r \geq 1$, there exists an open and dense set of $(f,\varphi) \in \mathcal{D}^r(M\times N,M)\times\mathcal{C}^r(M,\mathbb{R})$ for which the map $\Phi_{f,g,\varphi}$ is an embedding.

Note that, as already mentioned above, the set of g satisfying the conditions of this theorem is open and dense in $\mathcal{D}^r(N)$. It would be interesting to see whether this theorem generalizes to fully coupled systems, *i.e.* where g is not independent of x. Informally this would say that one can reconstruct coupled dynamical systems by observing a single subsystem. The appropriate conjecture would be that $\Phi_{f,g,\varphi}$ is an embedding for an open dense set of $((f,g),\varphi) \in \mathcal{D}^r(M\times N)\times \mathcal{C}^r(M,\mathbb{R})$. It seems likely that it should be possible to prove this using the techniques developed in [12], and work in this direction is in progress.

Theorem 3 shows that we can typically reconstruct the joint dynamics of a forcing and a forced system from a time series of an observable of the forced system alone. It is applicable in circumstances where the forcing is unknown, which will be the case in most practical applications. Sometimes, however, we may be in possession of independent knowledge of the state y of the forcing system. This will for instance be the situation if we control the forcing, as in many

laboratory experiments. If this is so, it seems that it should be unnecessary to have to reconstruct the forcing dynamics. Thus, rather than requiring $\Phi_{f,g,\varphi}$ to embed $M\times N$, we might simply ask that it should embed each $M\times\{y\}$. More precisely, if we define $\Phi_{f,g,\varphi,y} : M\times\{y\} \to \mathbb{R}^d$ by $\Phi_{f,g,\varphi,y}(x) = \Phi_{f,g,\varphi}(x,y)$ then we would want $\Phi_{f,g,\varphi,y}$ to be an embedding for each $y\in N$. We shall call such a $\Phi_{f,\varphi,y}$ a *bundle* or *fibre embedding*.

An embedding of this kind is for instance sufficient if we have independent knowledge of $y_i = g^i(y)$ and want to predict the time series $\varphi(x_i)$. Thus, as in §2 let $z_i = \Phi_{f,g,\varphi}(x_i,y_i)$. Then if for some d both Φ_{f,g,φ,y_i} and $\Phi_{f,g,\varphi,y_{i+1}}$ are embeddings of $M\times\{y_i\}$ and $M\times\{y_{i+1}\}$ respectively and we define $f_y : M\to M$ by $f_y(x) = f(x,y)$, then we have

$$z_{i+1} \quad = \quad \Phi_{f,g,\varphi,y_{i+1}}\circ f_{y_i}\circ(\Phi_{f,g,\varphi,y_i})^{-1}(z_i)$$

Thus if we let $G_{y_i} : \mathbb{R}^d \to \mathbb{R}$ be the last component of $\Phi_{f,g,\varphi,y_{i+1}}\circ f_{y_i}\circ(\Phi_{f,g,\varphi,y_i})^{-1}$, then

$$\varphi_{i+d} \quad = \quad G_{y_i}(\varphi_i, \varphi_{i+1}, \ldots, \varphi_{i+d-1})$$

where as usual $\varphi_i = \varphi(x_i)$. Then, in principle, given enough data we can estimate G_y as a function from $\mathbb{R}^d\times N$, and use this to predict the time series. The main benefit of this approach is that we do not waste resources in trying to reconstruct the dynamics of g, which we already know. In particular, since we are only trying to embed a m dimensional manifold, we would expect it to be sufficient to take $d \geq 2m+1$ rather than $d \geq 2(m+n) + 1$ as above. This in fact turns out to be the case. Since it is always desirable to work with the smallest d possible, this can be a significant advantage even if n is not particularly large. The one drawback is that unfortunately we are only able to show that $\Phi_{f,g,\varphi,y}$ is an embedding for a residual set of y, rather than for all $y\in N$. In particular the theorem we prove is:

Theorem 4 (Bundle Delay Embedding Theorem, Stark [12]) Let M and N be compact manifolds of dimension $m \geq 1$ and n respectively. Suppose that $d \geq 2m+1$ and the closure of the set of periodic orbits of period $\leq d$ of $g\in \mathcal{D}^r(N)$ is nowhere dense in N. Then for $r \geq 1$, there exists a residual set of $(f,\varphi)\in \mathcal{D}^r(M\times N,M)\times\mathcal{C}^r(M,\mathbb{R})$ such that for any (f,φ) in this set there is an open dense set of y such that $\Phi_{f,g,\varphi,y}$ is an embedding.

It remains an open question to what extent the conditions on g are necessary in this theorem. In particular, one might hope that it might be possible for $\Phi_{f,g,\varphi,y}$ to be generically an embedding for all g. There appears to be no obvious counterexample to this conjecture, and in particular the examples in [12] which show that conditions on g are necessary in Theorem 3 are irrelevant in the bundle embedding case since they fail to be embeddings only by mapping points with different y values to the same point. Similarly, one might ask whether it is possible for $\Phi_{f,g,\varphi,y}$ to be an embedding for all y, rather than just a residual set. Again, we have been unable to construct a counter-example to this. The only thing that we can say is that this can fail for an open set of (f,φ) when g has an open set of periodic orbits of short period, (see [12]).

To conclude this section, we want to discuss the implications of the results stated above to periodically forced differential equations. Recall that for such systems N is the unit circle $\mathbb{T}^1$ and g is given by $g(\theta) = \theta + \tau/T$, where τ is the sampling interval and T is the forcing period. This unfortunately means that if τ/T is a rational p/q then every point in $\mathbb{T}^1$ will be periodic with period q. Hence if $q \leq d$, then the conditions of Theorems 3 and 4 are not satisfied.

This is potentially a serious problem, since a very popular, and quite natural, choice is to sample at the forcing frequency, so that $\tau/T = 1$, $g = Id$, and every point in $\mathbb{T}^1$ is a fixed point. In such

a case [12], if the flow has a periodic orbit of period T which has a non-trivial projection onto M, then for an open set of φ, $\Phi_{f,g,\varphi}$ will fail to be 1-1, and hence fail to be an embedding of $M\times\mathbb{T}^1$.

Fortunately, in most practical applications this is not a great drawback. This is because when one takes $\tau = T$, one is usually thinking in terms of reconstructing the so called stroboscopic map for the system, which is in effect the Poincaré map for the section $\theta = \theta_0$ (for some constant θ_0). Thus, one is trying to embed $M\times\{\theta_0\}$ for a single θ_0, rather than the whole of $M\times\mathbb{T}^1$. In such a situation, we just have the single map f_{θ_0} to deal with (where $f_{\theta_0}(x) = f(x,\theta_0)$), and can justify the embedding procedure using the standard Takens Theorem. Thus for a residual set of f_{θ_0} and φ, the corresponding delay map (which is just $\Phi_{f,g,\varphi,\theta_0}$) will be an embedding. The same idea can be used when $\tau/T = p/q$, with $1 \le q < d$. Given θ_0, define $\theta_1, \ldots, \theta_{q-1}$ by $\theta_{i+1} = g(\theta_i)$. Then $\theta_0 = g(\theta_{q-1})$ and the orbit of θ_0 is restricted to $N = \{\theta_0, \ldots, \theta_{q-1}\}$ which is a zero dimensional manifold. Trivially, g has only a finite number of periodic points, and thus by Theorem 3 there is a dense open set of f such that $\Phi_{f,g,\varphi}$ embeds $M\times N = M\times\{\theta_0, \ldots, \theta_{q-1}\}$. Thus as long as we are satisfied with reconstructing the dynamics on $M\times\{\theta_0, \ldots, \theta_{q-1}\}$ only, Theorem 3 is perfectly adequate.

Recently, however, there has been considerable interest in embedding the whole of $M\times\mathbb{T}^1$, for instance when it is desired to reconstruct the braid structure of the periodic orbits (e.g. [13,14] and the references therein). In such cases it is usual to highly over-sample the time series, *i.e.* to take $\tau = T/q$, with q large, or τ/T irrational. By Theorem 3 the corresponding $\Phi_{f,g,\varphi}$ is generically again an embedding.

4. STOCHASTIC SYSTEMS

We now turn to the case of stochastic systems. Whilst our framework does not encompass all possible cases of stochastic dynamics (and in particular we only consider discrete time systems here), it includes a sufficiently rich class of systems to be both useful and interesting. We shall regard these as a deterministic system, forced by some stochastic process. This immediately makes the connection to the previous section obvious, and also allows us to treat input-output systems (see [15]), which are just deterministic systems forced by an arbitrary input sequence and irregularly sampled time series.

A standard approach to modelling systems forced by some non-deterministic process is through the use of shift spaces. Thus let N be some topological space, and define $\Sigma = N^{\mathbb{Z}}$ to be the space of bi-infinite sequences of elements in N with the product topology. Let $\sigma : \Sigma \to \Sigma$ be the standard shift map, thus $[\sigma(\omega)]_i = \omega_{i-1}$, where ω_i is the ith component of ω. Then if $f \in \mathcal{D}^r(M\times N, M)$ we get the skew product system

$$\begin{array}{lcl} x & \mapsto & f(x,\omega) \\ \omega & \mapsto & \sigma(\omega) \end{array}$$

Since the space Σ contains all possible sequences of elements in N, this gives us a very general model of systems driven by arbitrary sequences. Furthermore, if one wants to restrict interest to a particular class of input sequences, one can replace Σ by any shift invariant closed subset.

If in addition we have some measure μ on N then the corresponding product measure μ_Σ on Σ is shift invariant and hence (σ,μ_Σ) is a (Bernoulli) stochastic process. We can then think of f as a deterministic dynamical system driven by this process in the sense that we apply a different

map f_{ω_i} at each time step i, with ω_i chosen randomly with respect to μ at each time step. One often takes $N = \mathcal{D}^r(M)$ here, so that $f_{\omega_i}(x) = \omega_i(x)$. Systems of this kind are usually known as *random dynamical systems* or *random diffeomorphisms* (*e.g* [16,17]) and are widely used to model stochastic dynamical systems. It is also possible to consider more arbitrary shift invariant measures on Σ rather than just those given as product measures of a measure on N, and f could be taken to be a function f: $M \times \Sigma \to M$, rather than just depending on a single component ω_0 of ω. In such a case we write f_ω for the function given by $f_\omega(x) = f(x,\omega)$. For simplicity of exposition, however, we do not consider these generalizations here.

The natural concept of equivalence for two such systems f and f' is that of a *bundle conjugacy*, that is a map $h : M \times \Sigma \to M$ such that h_ω is a homeomorphism (or depending on the context, a diffeomorphism or measure isomorphism) for each ω, (where as usual $h_\omega(x) = h(x,\omega)$) and such that (h,Id) conjugates the skew dynamics of (f,Id) with the dynamics of (f',Id), *i.e.*

$$(f,\sigma)\circ(h,Id) \quad = \quad (h,Id)\circ(f',\sigma)$$

This can be seen to be equivalent to

$$f_\omega \circ h_\omega \quad = \quad h_{\sigma(\omega)} \circ f'_\omega$$

and is analogous to the concept of bundle embedding in §3. Ideally we would like $f_\omega \circ h_\omega = h_{\sigma(\omega)} \circ f'_\omega$ to hold for every $\omega \in \Sigma$, but in general this is too ambitious, and we have be content with a conjugacy only for "typical" ω, i.e. for generic ω in the topological setting, or for almost every ω with respect to μ_Σ in the measure theoretic context. Note that even when, as in our case, f depends only on ω_0 we still allow h to depend on the other components of ω.

Now suppose that we have such a random dynamical system and we observe it using some function $\varphi : M \to \mathbb{R}$. If we define $f_{\omega_i \ldots \omega_0} = f_{\omega_i} \circ \ldots \circ f_{\omega_0}$, where $f_{\omega_i}(x) = f(x,\omega_i)$, then the usual delay embedding can then be written

$$\Phi_{f,\varphi,\omega}(x) \quad = \quad (\varphi(x), \varphi(f_{\omega_0}(x)), \ldots, \varphi(f_{\omega_{d-2} \ldots \omega_0}(x)))$$

By analogy to Theorem 4 above, if we wish to reconstruct the dynamics of our system using delay co-ordinates, then it is reasonable to require $\Phi_{f,\varphi}$ to be a bundle embedding, *i.e.* $\Phi_{f,\varphi,\omega}$ should be an embedding (in the usual sense) for "typical" ω. Recall that in Theorem 4 we made no attempt to reconstruct the forcing dynamics g, and as a consequence, the embedding dimension d was independent of the dimension n of the forcing system. Since the only difference between the setting here, and that of Theorem 4, is that we have replaced the finite dimensional dynamics g by the infinite dimensional shift map σ, it is not unreasonable to expect that we should still get a bundle embedding here. In fact, it turns out to be a straightforward matter to modify Theorem 4 to give

Theorem 5 (Stark, Broomhead, Davies and Huke, [18]) Let M and N be compact manifolds of dimension $m \geq 1$ and n respectively. Suppose that $d \geq 2m+1$. Then for $r \geq 1$, there exists a residual set of $(f,\varphi) \in \mathcal{D}^r(M \times N, M) \times \mathcal{C}^r(M,\mathbb{R})$ such that for any (f,φ) in this set there is an open dense set of ω in Σ such that $\Phi_{f,\varphi,\omega}$ is an embedding.

Note that in contrast to the general discussion of random dynamical systems above, we take N to be a finite dimensional compact manifold here, some such restrictions is necessary in order to give us some control of the topology on $\mathcal{D}^r(M \times N, M)$. One can also give a measure theoretic version of Theorem 5, which leads to a Takens Embedding Theorem for Stochastic Systems.

Ideally, we would like to take $N = \mathcal{D}^r(M)$ in this case, with μ an arbitrary measure $\mathcal{D}^r(M)$ and obtain an embedding for μ_Σ-almost every ω. However, this clearly cannot hold in general, since we could simply take μ to be the point measure on a $f \in \mathcal{D}^r(M)$ which does not satisfy the conditions of Theorem 2 (*e.g.* $f = Id$ will do). We therefore have to restrict N and μ in some way. A reasonable choice, which allows us to use the same techniques as in the proof of Theorems 4 and 5, is to require N to be a compact finite dimensional submanifold of $\mathcal{D}^r(M)$, and μ to be absolutely continuous with respect to Lebesgue measure on N. Then μ_Σ corresponds to an *i.i.d.* random process with an absolutely continuous distribution. We can think of N as defining a finite parameter family of diffeomorphisms of M, so that our notion of a stochastic dynamical system is that of a finite parameter family of maps with *i.i.d.* noise on the parameters. This would seem to include a sufficiently rich class of systems to encompass many real applications.

As usual, we can only expect to get an embedding for "typical" submanifolds N of $\mathcal{D}^r(M)$; thus as above if we took $N = \{Id\}$, then $\Phi_{f,\varphi}$ would not be a bundle embedding for generic φ. Before we can state our Stochastic Embedding Theorem it therefore remains to give a rigorous definition of the concept of a "typical" submanifold of $\mathcal{D}^r(M)$. Note that $\mathcal{C}^r(N,\mathcal{D}^r(M))$ is a Banach manifold (e.g. [12, 19]) and hence we have the usual notion of a generic $\rho \in \mathcal{C}^r(N,\mathcal{D}^r(M)$. Observe that ρ associates a diffeomorphism of M with each point of the finite dimensional manifold N. We thus think of a "typical" submanifold of $\mathcal{D}^r(M)$ as the image $\rho(N)$ of a fixed N for ρ belonging to a generic subset of $\mathcal{C}^r(N,\mathcal{D}^r(M)$. This is equivalent to the concept of finite parameter family of diffeomorphisms of M with generic dependence on the parameters. Given $\rho \in \mathcal{C}^r(N,\mathcal{D}^r(M)$, define $f_{\omega_i} = \rho(\omega_i)$, and let $f_{\omega_i \ldots \omega_0} = f_{\omega_i} \circ \ldots \circ f_{\omega_0}$ as before. Then for $(\rho,\varphi) \in \mathcal{C}^r(N,\mathcal{D}^r(M)) \times \mathcal{C}^r(M,\mathbb{R})$ we define a delay map $\Phi_{\rho,\varphi,\omega}$ exactly as above by

$$\Phi_{\rho,\varphi,\omega}(x) \quad = \quad (\varphi(x), \varphi(f_{\omega_0}(x)), \ldots, \varphi(f_{\omega_{d-2} \cdots \omega_0}(x)))$$

Our Takens Embedding Theorem for Stochastic Systems is then given by:

Theorem 6 (Stark, Broomhead, Davies and Huke, [18]) Let M and N be compact manifolds of dimension $m \geq 1$ and n respectively. Suppose that $d \geq 2m+1$. Then for $r \geq 1$, there exists a residual set of $(\rho,\varphi) \in \mathcal{C}^r(N,\mathcal{D}^r(M)) \times \mathcal{C}^r(M,\mathbb{R})$ such that for any (ρ,φ) in this set there is an open dense set $\Sigma_{\rho,\varphi} \subset \Sigma$ such that $\Phi_{\rho,\varphi,\omega}$ is an embedding for all $\omega \in \Sigma_{\rho,\varphi}$. Furthermore if μ is a measure on N which is absolutely continuous with respect to Lebesgue measure and μ_Σ the corresponding product measure on Σ, then we can choose $\Sigma_{\rho,\varphi}$ such that $\mu_\Sigma(\Sigma_{\rho,\varphi}) = 1$.

It is also straightforward to incorporate noise in the observations, so that φ also depends on ω [18] (or possibly on an independent process ω'). The first part of this theorem gives Theorem 5 as an immediate corollary since the map $\mathcal{H} : \mathcal{C}^r(N,\mathcal{D}^r(M)) \to \mathcal{D}^r(M \times N,M)$ defined by $\mathcal{H}(\rho)(x,y) = \rho(y)(x)$ is a homeomorphism and by definition

$$\begin{aligned}
\Phi_{\rho,\varphi,\omega}(x) \quad &= \quad (\varphi(x), \varphi(\rho(\omega_0)(x)), \ldots, \varphi(\rho(\omega_{d-2}) \circ \ldots \circ \rho(\omega_0)(x))) \\
&= \quad (\varphi(x), \varphi(\mathcal{H}(\rho)(x,\omega_0)), \ldots, \varphi(\mathcal{H}(\rho)(x,\omega_{d-2}) \circ \ldots \circ \mathcal{H}(\rho)(x,\omega_0)(x))) \\
&= \quad \Phi_{\mathcal{H}(\rho),\varphi,\omega}
\end{aligned}$$

Similarly, we can give an alternative formulation of the second part of Theorem 6:

Theorem 7 Let M and N be compact manifolds of dimension $m \geq 1$ and n respectively and let μ be a measure on N which is absolutely continuous with respect to Lebesgue measure. Suppose that $d \geq 2m+1$. Then for $r \geq 1$, there exists a residual set of $(f,\varphi) \in \mathcal{D}^r(M \times N,M) \times \mathcal{C}^r(M,\mathbb{R})$ such that for any (f,φ) in this set $\Phi_{f,\varphi,y}$ is an embedding for almost every y (with respect to the product measure on Σ).

To conclude the paper, we discuss the implications of Theorems 5-7 for the delay reconstruction of time series, these are virtually identical to those for Theorem 4. Thus, using the notation of Theorems 5 and 7, suppose that ω is such that $\Phi_{f,\varphi,\omega}$ and $\Phi_{f,\varphi,\sigma(\omega)}$ are both embeddings of M. Then the map $F_\omega = \Phi_{f,\varphi,\sigma(\omega)} \circ f_{\omega_0} \circ (\Phi_{f,\varphi,\omega})^{-1}$ is well defined and is a diffeomorphism between $\Phi_{f,\varphi,\omega}(M) \subset \mathbb{R}^d$ and $\Phi_{f,\varphi,\sigma(\omega)}(M) \subset \mathbb{R}^d$. Let $(x_i, \sigma^i(\omega))$ be an orbit of (f,σ), so that $x_{i+1} = f(x_i, \omega_i)$, and as usual define $\varphi_i = \varphi(x_i)$ and $z_i = (\varphi_i, \varphi_{i+1}, \ldots, \varphi_{i+d-1})$. Then $z_i = \Phi_{f,\varphi,\sigma^i(\omega)}(x_i)$ and hence if $\Phi_{f,\varphi,\sigma^i(\omega)}$ and $\Phi_{f,\varphi,\sigma^{i+1}(\omega)}$ we have

$$\begin{aligned} z_{i+1} &= \Phi_{f,\varphi,\sigma^{i+1}(\omega)}(x_{i+1}) \\ &= \Phi_{f,\varphi,\sigma^{i+1}(\omega)}(f_{\omega_i}(x_i)) \\ &= \Phi_{f,\varphi,\sigma^i(\sigma(\omega))}(f_{\omega_i}((\Phi_{f,\varphi,\sigma^i(\omega)})^{-1}(z_i))) \\ &= F_{\sigma^i(\omega)}(z_i) \end{aligned}$$

Therefore in exact analogy to the standard Takens framework, $F_{\sigma^i(\omega)}$ is just the map which shifts a block of the time series forward by one time step, and hence $(\varphi_i, \varphi_{i+1}, \ldots, \varphi_{i+d-1}) \mapsto (\varphi_i, \varphi_{i+1}, \ldots, \varphi_{i+d-1})$ is bundle conjugate to our original dynamics f_{ω_i}. Note however, that whereas f_{ω_i} only depends on $\omega_i = \sigma^i(\omega)_0$, the map $F_{\sigma^i(\omega)}$ depends on $\omega_i, \omega_{i+1}, \ldots, \omega_{i+d-1}$.

The first d-1 components of F_ω are trivial. If we denote the last component by $G_\omega : \Phi_{f,\varphi,\omega}(M) \to \mathbb{R}$ then

$$\varphi_{i+d} = G_{\sigma^i(\omega)}(\varphi_i, \varphi_{i+1}, \ldots, \varphi_{i+d-1})$$

If we write out the dependence on $\sigma^i(\omega)$ explicitly, we get

$$\varphi_{i+d} = G(\varphi_i, \varphi_{i+1}, \ldots, \varphi_{i+d-1}, \omega_i, \omega_{i+1}, \ldots, \omega_{i+d-1})$$

In the case of one dimensional ω_i the existence of such a function was conjectured by Casdagli [15]. From another point of view, processes which satisfy this equation (again for a one dimensional ω_i) are well known in signal processing under the name of Nonlinear Auto-Regressive Moving Average (NARMA) models (e.g. [20]). Note however, that in the case of NARMA processes, one usually assumes that G depends smoothly on all its arguments, whilst we can say nothing about the regularity of its dependence on $\omega_i, \omega_{i+1}, \ldots, \omega_{i+d-1}$, indeed G is only defined for almost every $\omega_i, \omega_{i+1}, \ldots, \omega_{i+d-1}$ with respect to the product measure μ_d on N^d.

ACKNOWLEDGEMENTS

We would like to thank Ludwig Arnold and Colin Sparrow for a number of useful conversations which were crucial to the development of the research presented here. Some of the preliminary work described here was carried out at IMPA, Brazil, during the International Conference on Dynamical Systems held there in August 1993, and JS is extremely grateful to IMPA for its hospitality and financial support, and to the Royal Society for providing travel funds to attend this conference. Part of this work was carried out at the Basic Research Institute in Mathematical Sciences at Hewlett-Packard in Bristol in the summer of 1994 and at the Newton Institute programme "From Finite to Infinite Dimensional Dynamical Systems" during the summer of 1995 and JS would like to thank both Institutes for their hospitality and financial support. Finally, JS would like to gratefully acknowledge financial support from the Nuffield Foundation (SCI/180/92/365/G), MED would like to thank the Royal Society for financial support and both JS and MED acknowledge financial support from the UK Engineering and Physical Sciences Research Council (GR/J97243) and

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